

Lecture 5: Newton's Method: Local Convergence

Math 6610

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Today's Goals

- Extend Newton's method to systems via the Jacobian

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- Recognize that each Newton step is one linear solve (LU!)
- Prove affine invariance and understand why it matters
- Preview the convergence theorem and the norms it needs

Newton's Method for $n \geq 1$: Setup

Remark (*Newton's Method for $n \geq 1$*). n equations, n unknowns:

$$f : D \rightarrow \mathbb{R}^n, \quad D \subset \mathbb{R}^n, \quad f = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in D.$$

Goal: find $x \in D$ with $f(x) = 0$. Initial guess: $x^0 \in D$. Taylor's theorem at x^0 :

$$f(x) = f(x^0) + Df(x^0)(x - x^0) + O(\|x - x^0\|^2) \quad (1)$$

where

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} = \text{Euclidean norm.}$$

The Jacobian and the Taylor Remainder

Definition (Jacobian).

$$Df(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \left[\frac{\partial f_j}{\partial x_k} \right]_{j,k=1,2,\dots,n} = \text{Jacobian of } f,$$

the $n \times n$ matrix of first partial derivatives; row j belongs to f_j .

Remark (The remainder in (1)). $O(\|x - x^0\|^2)$ is a term $R(x)$ with

$$\|R(x)\| \leq \text{const} \cdot \|x - x^0\|^2 \quad \text{for all } x \text{ near } x^0.$$

This holds when Df is Lipschitz continuous near x^0 (e.g. f twice continuously differentiable). Near x^0 the remainder is much smaller than the two terms kept.

One Newton Step

Remark (One Newton step). Drop the remainder in (1) and set the rest to 0:

$$0 = f(x^0) + Df(x^0) \underbrace{(x^1 - x^0)}_{\Delta x^0} \iff \begin{cases} Df(x^0) \Delta x^0 = -f(x^0) \\ x^1 = x^0 + \Delta x^0. \end{cases}$$

n linear equations for the n entries of Δx^0 ; exactly one solution $\Leftrightarrow Df(x^0)$ nonsingular (invertible). Repeat from x^1 .

Remark (Linear algebra inside). Each step k : a linear system with matrix $A = Df(x^k) \Rightarrow$ LU factorization of A (Lectures 1–3). Never form $Df(x^k)^{-1}$.

Newton's Method: Algorithm

Remark (Newton's Method Algorithm). Choose $x^0 \in D$.

- for $k = 0, 1, 2, \dots$ (open-ended), do:
 - ▶ if $Df(x^k)$ is singular, stop. (No unique Δx^k .)
 - ▶ else, solve $Df(x^k) \Delta x^k = -f(x^k)$ for Δx^k
 - ▶ set $x^{k+1} = x^k + \Delta x^k$
 - ▶ if $x^{k+1} \notin D$, stop. (f undefined there.)
 - ▶ check for convergence: if $f(x^{k+1}) \approx 0$, stop.
- end (k)

Affine Invariance: Definition

Definition (Affine invariance). Let $M \in \mathbb{R}^{n \times n}$ be a fixed nonsingular matrix and

$$g : D \rightarrow \mathbb{R}^n, \quad g(x) := Mf(x) \quad \text{for all } x \in D.$$

(g = the equations of f , rescaled and mixed with fixed coefficients.) Since M is invertible,

$$f(x) = 0 \Leftrightarrow Mf(x) = 0 \Leftrightarrow g(x) = 0,$$

so f and g have the same zeros. An iterative method is called *affine-invariant* if, for every such M and every $x^0 \in D$, it produces the same iterates x^k , $k = 0, 1, 2, \dots$, for f and for g .

Proposition. Newton's method is affine-invariant. (It also stops for a singular Jacobian at the same step, if at all.)

Affine Invariance: Proof (1/2)

Proof. Induction on k : both iterations start at x^0 ; assume both are at x^k .

1. *Jacobian of g .* $g_j(x) = \sum_l M_{jl} f_l(x)$ with constant M_{jl} , so

$$\frac{\partial g_j}{\partial x_m} = \sum_{l=1}^n M_{jl} \frac{\partial f_l}{\partial x_m} = [M Df(x)]_{jm} \quad \implies \quad Dg(x) = M Df(x) \quad \text{on } D.$$

2. *Singularity test.* $\det Dg(x^k) = \det M \cdot \det Df(x^k)$ and $\det M \neq 0$, so

$$Dg(x^k) \text{ singular} \Leftrightarrow Df(x^k) \text{ singular} :$$

both iterations stop here, or neither does.

(continued on the next slide)

Affine Invariance: Proof (2/2)

Proof (continued). 3. Corrections. Let $Df(x^k)$ be nonsingular. Multiplying on the left by M , resp. M^{-1} ,

$$Dg(x^k) \Delta x^k = -g(x^k) \Leftrightarrow MDf(x^k) \Delta x^k = -Mf(x^k) \Leftrightarrow Df(x^k) \Delta x^k = -f(x^k).$$

Both systems have a nonsingular matrix, hence exactly one solution each, and these solutions coincide. So both iterations get the same Δx^k , the same $x^{k+1} = x^k + \Delta x^k$, and the same answer to “ $x^{k+1} \notin D$ ”. ■

Remark (Not invariant: the residual test). $g(x^{k+1}) = Mf(x^{k+1})$ can be made as large or small as we like by the choice of M , while the iterates do not change.

Convergence of Newton's Method: What to Expect

Proposition (Convergence). Let x^* be a zero of f , i.e. $f(x^*) = 0$, with $Df(x^*)$ nonsingular and Df Lipschitz continuous near x^* (e.g. f twice continuously differentiable). Then there is a ball centered at x^* such that for every x^0 in it: all $Df(x^k)$ are nonsingular, all x^k stay in the ball,

$$\lim_{k \rightarrow \infty} x^k = x^*,$$

and the convergence is quadratic: for some constant C ,

$$\|x^{k+1} - x^*\| \leq C \|x^k - x^*\|^2, \quad k = 0, 1, 2, \dots$$

Proof: Lecture 6.

Quadratic Convergence

Definition (Quadratic Convergence). $x^k \rightarrow x^*$ converges *quadratically* if there is $C \geq 0$ with

$$\|x^{k+1} - x^*\| \leq C \|x^k - x^*\|^2 \quad \text{for all } k = 0, 1, 2, \dots$$

Remark (Correct digits roughly double). Multiply the inequality by C :

$$C \|x^{k+1} - x^*\| \leq (C \|x^k - x^*\|)^2.$$

Hence

$$C \|x^k - x^*\| \leq 10^{-m} \implies C \|x^{k+1} - x^*\| \leq 10^{-2m} :$$

about m correct digits at step k give about $2m$ at step $k + 1$.

The Convergence Theorem (Preview)

Theorem (Convergence of Newton's Method, Affine-Invariant). Assumptions:

- $D \subset \mathbb{R}^n$ is convex ($x, y \in D \Rightarrow$ the segment from x to y lies in D)
- $f : D \rightarrow \mathbb{R}^n$ is C^1 (all $\frac{\partial f_j}{\partial x_k}$ exist and are continuous on D)
- $x^0 \in D$ and $Df(x^0)$ is nonsingular
- Df satisfies the affine-invariant Lipschitz condition: for some $\gamma \geq 0$,

$$\|(Df(x^0))^{-1}(Df(y) - Df(x))\| \leq \gamma \|y - x\| \quad \text{for all } x, y \in D$$

Conclusion (more details in Lecture 6): if $\|\Delta x^0\|$ is small enough compared with $1/\gamma$ and D contains a suitable ball around x^0 , then all x^k are well defined, stay in that ball, and converge quadratically to a zero x^* of f .

The Norms in the Theorem

Definition (Vector norm, associated matrix norm). $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *vector norm* if for all $x, y \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$:

$$\|x\| \geq 0, \quad \|x\| = 0 \Leftrightarrow x = 0; \quad \|\alpha x\| = |\alpha| \|x\|; \quad \|x + y\| \leq \|x\| + \|y\|.$$

(The theorem holds for any vector norm.) The *associated matrix norm* is

$$\|M\| := \max_{\|x\|=1} \|Mx\|, \quad M \in \mathbb{R}^{n \times n}.$$

The maximum exists: $x \mapsto \|Mx\|$ is continuous and $\{x : \|x\| = 1\}$ is closed and bounded.

Properties of the Matrix Norm

Lemma. For all $M, N \in \mathbb{R}^{n \times n}$ and $x \in \mathbb{R}^n$:

- ❶ $\|Mx\| \leq \|M\| \|x\|$
- ❷ $\|I\| = 1$
- ❸ $\|MN\| \leq \|M\| \|N\|$

Proof. 1. For $x = 0$ both sides are 0. For $x \neq 0$, $\|x/\|x\|\| = 1$, so

$$\frac{\|Mx\|}{\|x\|} = \left\| M \frac{x}{\|x\|} \right\| \leq \|M\|.$$

2. $\|Ix\| = \|x\| = 1$ for every x with $\|x\| = 1$, so the maximum is 1.

3. For $\|x\| = 1$, by 1. twice:

$$\|MNx\| = \|M(Nx)\| \leq \|M\| \|Nx\| \leq \|M\| \|N\| \|x\| = \|M\| \|N\|.$$

Take the maximum over such x . ■

Filling in the Details: One Newton Step, Concretely

Take the 2×2 system

$$f(x) = \begin{bmatrix} x_1^2 + x_2^2 - 1 \\ x_2 - x_1^2 \end{bmatrix} = 0 \quad (\text{circle} \cap \text{parabola}).$$

Jacobian:

$$Df(x) = \begin{bmatrix} 2x_1 & 2x_2 \\ -2x_1 & 1 \end{bmatrix}.$$

From $x^0 = (1, 1)^T$: $f(x^0) = (1, 0)^T$ and

$$\begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix} \Delta x^0 = - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \Delta x^0 = \begin{bmatrix} -1/6 \\ -1/3 \end{bmatrix}, \quad x^1 = \begin{bmatrix} 5/6 \\ 2/3 \end{bmatrix}.$$

(Row 2: $\Delta x_2^0 = 2\Delta x_1^0$; row 1 then gives $6\Delta x_1^0 = -1$.) Note the workflow: evaluate f , evaluate Df , **LU-solve**, update. Never form $Df(x^k)^{-1}$ explicitly – solving the system is cheaper and more stable.

Computational Cost per Newton Step

Cost per Newton step for a dense Jacobian:

$$\underbrace{n^2 \text{ entries of } Df}_{\text{evaluation}} + \underbrace{\frac{2n^3}{3}}_{\text{LU}} + \underbrace{2n^2}_{\text{triangular solves}} .$$

The LU factorization dominates.

Why Affine Invariance Is the Right Point of View

$g := Mf$ with M nonsingular: same solution set, and (by the proof) the same Newton iterates:

$$Dg(x^k)^{-1}g(x^k) = (MDf(x^k))^{-1}Mf(x^k) = Df(x^k)^{-1}M^{-1}Mf(x^k) = Df(x^k)^{-1}f(x^k).$$

Consequences:

- Any *theory* for Newton's method should be invariant under $f \mapsto Mf$. The raw condition $\|Df(y) - Df(x)\| \leq \gamma \|y - x\|$ is not (its left side becomes $\|M(Df(y) - Df(x))\|$); the one in the theorem is:

$$\begin{aligned} Dg(x^0)^{-1}(Dg(y) - Dg(x)) &= Df(x^0)^{-1}M^{-1}M(Df(y) - Df(x)) \\ &= Df(x^0)^{-1}(Df(y) - Df(x)), \end{aligned}$$

so the same γ works for f and g .

- Any *convergence test* should be invariant too. Checking $\|f(x^k)\|$ is not (multiply f by 10^6 !); checking the Newton correction $\|Df(x^k)^{-1}f(x^k)\| = \|\Delta x^k\|$ is, since Δx^k is the same for f and g . This foreshadows the monotonicity test of Lecture 7.

Looking Ahead

- **Next lecture:** The Newton Convergence Theorem